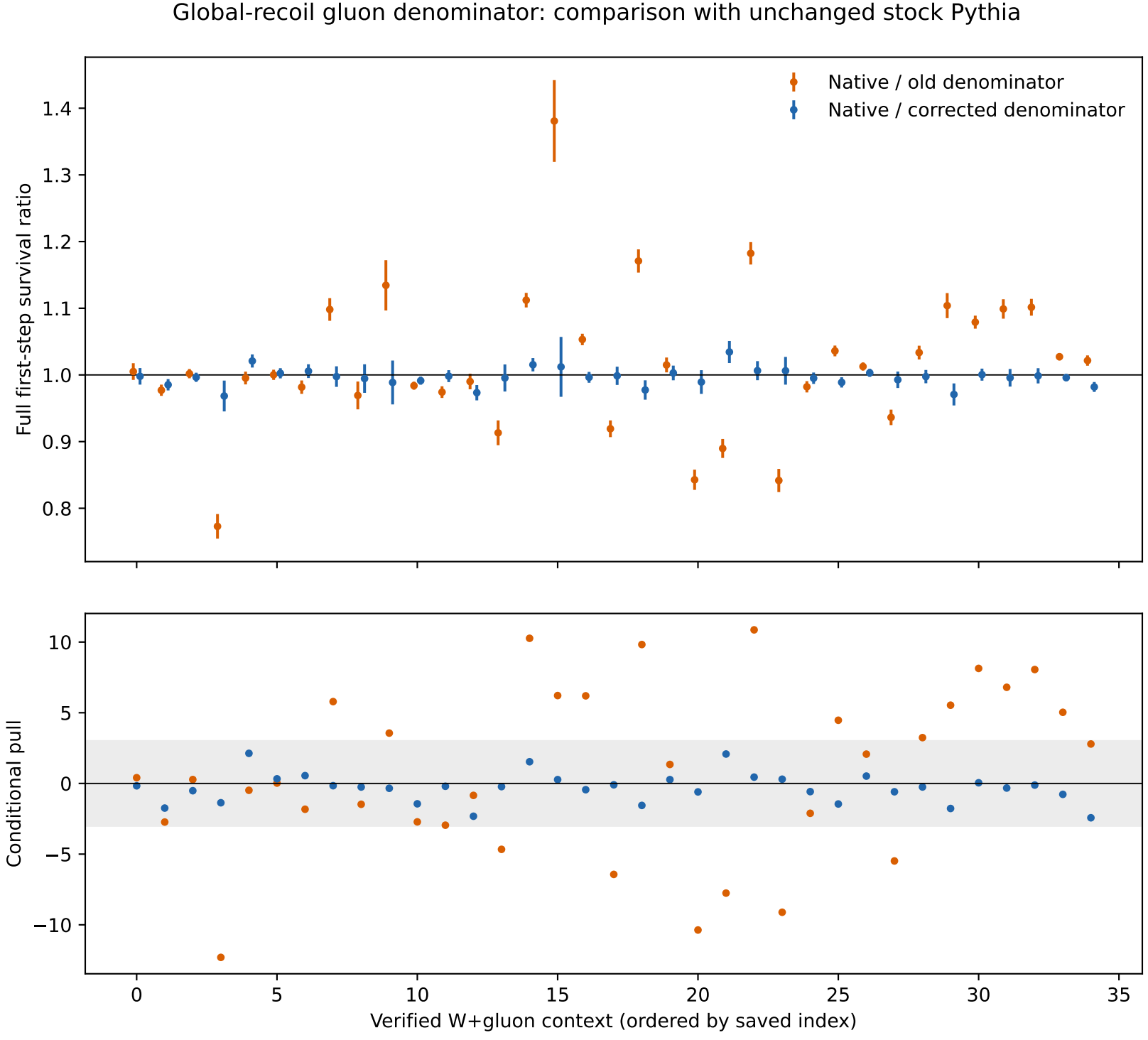
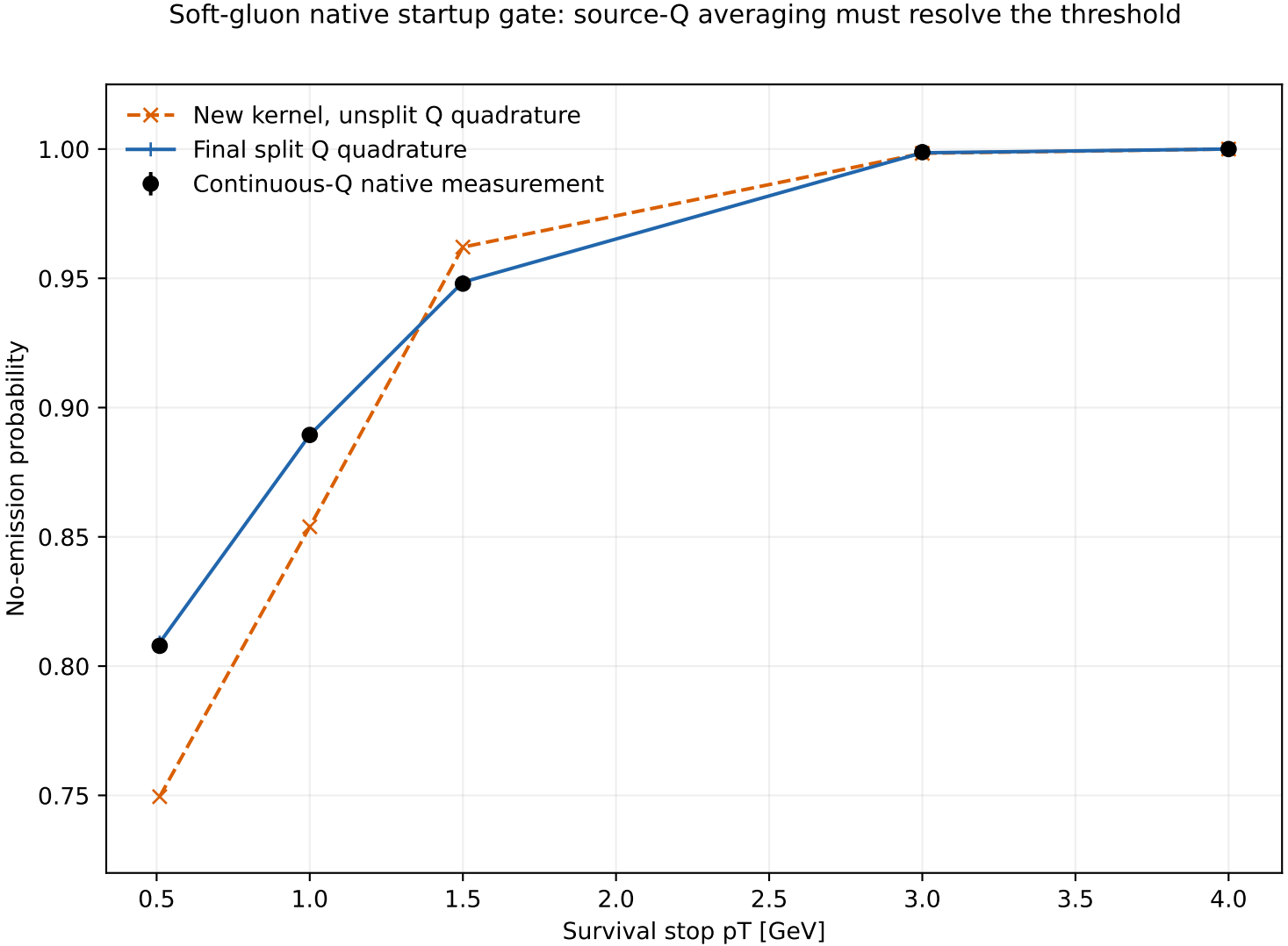

Global-recoil gluon denominator: comparison with unchanged stock Pythia
Native / old denominator
Native / corrected denominator
Full first-step survival ratio
1.4
1.3
1.2
1.1
1.0
0.9
0.8
Conditional pull
10
5
0
−5
−10
0
5
10
15
20
25
30
35
Verified W+gluon context (ordered by saved index)
35 selected contexts, 5,000 native trials each. Conditional binomial errors; no population residual extrapolation.


Soft-gluon native startup gate: source-Q averaging must resolve the threshold
New kernel, unsplit Q quadrature
Final split Q quadrature
Continuous-Q native measurement
No-emission probability
1.00
0.95
0.90
0.85
0.80
0.75
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Survival stop pT [GeV]
Physical W+g Born, gluon energy 7 GeV; starting-Q prior [1,10] GeV; 50,000 independent trials per stop.